

Nirmit K. Tripathii

AI / ML ENGINEER & RESEARCHER

Production-grade AI systems — multi-agent LLM orchestration, multimodal RAG, speech & video ML, legal AI, and civic tech. M.Tech in Modeling & Simulation with 3+ years shipping from ideation to deployment on real-world (often free-tier) hardware.

Email nirmitktripathii@yahoo.com **Location** Delhi NCR, India · Remote **Portfolio** nirmitktripathii.github.io/PortfolioSite
GitHub github.com/nirmitktripathii **LinkedIn** in/nirmitk-tripathii

AVAILABILITY

Open to freelance, consulting & research collaboration — remote. Open to full-time from Sept 2026.

CORE SKILLS

Agentic AI & LLMs

LangGraph LangChain LangSmith
MCP RAG · LightRAG · CRAG
VideoRAG FAISS

Languages & Frameworks

Python TypeScript / Node PyTorch
FastAPI Flask SQL

Models & APIs

Gemini OpenAI Llama 3.1
HuggingFace IndicF5 XLM-RoBERTa
IndicBERT InLegalBERT ImageBind
Whisper

Data & MLOps

PostgreSQL Neo4j NetworkX
HNSWLib Docker W&B OpenCV
Azure OCR Pandas · NumPy

Sports AI

xG Modelling VAEP FUTR
T-DEED SoccerNet

EDUCATION

M.Tech — Modeling & Simulation

Savitribai Phule Pune University
Sep 2019 - Feb 2022

B.Tech — Electrical Engineering

Pandit Deendayal Petroleum University
Jul 2014 - Jun 2018

CERTIFICATIONS

Masters Program — Data Scientist

Simplilearn

Football Data Analytics

MAS About Sports

WRITING

11 engineering deep-dives — dubbing fine-tuning, multi-agent MCP infrastructure, resilient zero-cost video generation, SoccerNet anticipation, civic-tech systems & agri-policy research. On the portfolio.

EXPERIENCE

AI/ML Software Engineer · Chetu India

Apr 2025 - Jun 2026 · Noida

- Built **DIGEST** — an agentic LangGraph (StateGraph) chatbot for event discovery & ticket commerce, orchestrating multi-step workflows across 7 intents with the full Automatiq B2B transaction lifecycle.
- Architected a **Video-to-SOP** multimodal pipeline: ImageBind embeddings + a Neo4j/NetworkX VideoRAG knowledge graph, a LangGraph refinement loop, and LangSmith observability.
- Built **IKS Oracle (Gyanstrot)** — a RAG system over Indian Knowledge Systems, deployed live on HuggingFace Spaces.

AI/ML Engineer · Blu Parrot

Jan 2024 - Apr 2025 · Gurgaon

- LegalGPT** — full RAG pipeline for Indian legal research & compliance; fine-tuned InLegalBERT/InCaseLawBERT across 150+ contract types; compared Naive/Light/CRAG retrieval.
- Email Orchestrator** — a multi-account MCP email agent consolidating four providers behind one MCP server that is itself an MCP client; 17 unified tools, per-account fan-out resilience, 49 passing tests.
- Hindi ASR portal on NeMo Conformer-CTC (**92%+** zero-shot) with Gemini summarisation; a 22-language Indic translation portal (IndicBERT + IndicTrans2, **94%** zero-shot).
- Promoted to **Project Lead** for a UK real-estate multimodal appraisal regression model.

Subject Matter Expert, AI/ML · House of Coutons

Nov 2023 - Jan 2024

- Delivered 3 research papers and a dissertation across literature review, EDA, model building and inference.

AI/ML Programmer · HRBOTICS (Credence Maven)

Apr 2022 - Mar 2023

- Multi-threaded Python scrapers for NCR commercial data; automated LinkedIn hiring workflows (Selenium + BeautifulSoup) with a reusable ATS prototype.

Football Data Analyst · Boomer 11

Nov 2021 - Mar 2022

- Built an Expected Goals (xG) model (RF Classifier) at **89.77%** accuracy — published research; delivered player-rating and real-time performance analysis.

SELECTED PROJECTS

Indic AI Dubbing Platform · Llama-3.1-8B QLoRA · IndicF5 · Gemini · Faster-Whisper · Demucs
Isochrony-aware dubbing across 11 Indic languages. The v3 fine-tune constrains translation to a phoneme budget, cutting length error **79%** (0.495→0.103 rel.) while *improving* quality (chrF++ 22.0→30.1); deployed v2 uses Gemini CoT + IndicF5 zero-shot cloning, running on a free Kaggle T4.

FAANTRA — SoccerNet 2026 Ball Action Anticipation · PyTorch · FUTR · T-DEED · RegNetY-008
Predicts ball actions seconds ahead from broadcast video with decoupled what/when heads. Official challenge-server **mAP_Average 17.69** (mAP@∞ 23.59, mAP@5 22.47) — the ∞-vs-1 gap shows the model reads *what* far better than *when*.

India Aaspaas — Civic Reporting Platform · Node/Express · PostgreSQL · React · Cloudflare R2
Production map-based civic-issue platform with server-rendered social share cards, a two-tier trust model, and a hardened upload path — 67 tests against a real Postgres engine, ~\$3/month to run.

Podcast Shorts Factory · Multi-agent · Python · open source (MIT)
Ten cooperating AI agents that turn long podcasts into scheduled short-form videos — provider-agnostic LLM layer with automatic fallback, a learning feedback loop, and a QA gate; runs unattended on free providers.

AI Video Style Cloner · Streamlit · Multi-provider LLM failover · Edge-TTS · MoviePy
Paste a YouTube channel URL → it reverse-engineers the house style, writes an original script in that voice, then narrates, illustrates & compiles a finished MP4 for **\$0**. A three-tier LLM failover, self-healing JSON validators, and a zero-network procedural fallback guarantee output even when every API is down.

Football Analytics — VAEP & Expected Goals · RF · SVR · Streamlit
VAEP action-valuation dashboard across Top-5 leagues & World Cup 2018; published xG thesis (RF Classifier, 89.77% / 90.55% post-HPO).